



FAST NUCES Karachi

Kinexis: *Biomechanical Feedback & Error Detection System via Human Pose Estimation*

Deep Learning for Perception
Project Report

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Submission date:

10th May 2026

1. INTRODUCTION

Human movement analysis is important in areas such as physiotherapy, fitness, rehabilitation, and sports training. Incorrect exercise posture or movement patterns can lead to injuries and reduced performance. Traditionally, movement assessment depends on manual observation by trainers or medical experts, which can be time-consuming and difficult to access regularly. Recent advances in computer vision and deep learning have enabled automated systems that can analyze human motion directly from video data.

This project presents a skeleton-based human action recognition system using a fine-tuned ST-GCN++ (Spatial Temporal Graph Convolutional Network) model. The system uses human pose keypoints extracted from video sequences to learn both spatial relationships between body joints and temporal movement patterns across frames. A pretrained ST-GCN++ backbone is adapted through transfer learning to improve performance on exercise action recognition tasks.

The model is trained to classify multiple exercise actions including pull-ups, push-ups, bench press, jumping jacks, sit-ups, jump rope, and squats.

1.1. Objectives

- To develop a real-time skeleton-based human action recognition system for exercise analysis using deep learning.
- To fine-tune a pretrained ST-GCN++ model for classifying exercise movements from human pose keypoints.
- To analyze spatial joint relationships and temporal motion patterns using graph-based learning.
- To integrate repetition counting, biomechanical feedback, and confidence-based pose analysis into a unified pipeline.
- To evaluate the effectiveness of transfer learning and system performance using standard classification metrics on the Penn Action Dataset.

2. METHODOLOGY

2.1 System Overview

The proposed system is a skeleton-based human action recognition pipeline designed to classify exercise movements using deep learning. The pipeline consists of three main stages: preprocessing, skeleton-based feature modeling, and classification.

First, human pose keypoints are extracted from video sequences and converted into a structured skeleton representation. These keypoints are then processed by a fine-tuned ST-GCN++ (Spatial Temporal Graph Convolutional Network) model, which learns both spatial relationships between body joints and temporal motion dynamics across frames. Finally, the trained model predicts the exercise category performed in the video.

The system is trained and evaluated on exercise-based action classes including pull-ups, push-ups, bench press, jumping jacks, sit-ups, jump rope, and squats.

2.2 Data Preprocessing

The preprocessing stage prepares the skeletal data for deep learning training and inference.

2.2.1 Skeleton Representation

Each video sequence is represented using 2D human pose keypoints. The skeleton is modeled as a graph where:

- Joints represent graph nodes
- Bone connections represent graph edges

The selected skeleton contains 13 major body joints connected using a predefined adjacency structure.

2.2.2 Frame Sampling and Sequence Processing

To ensure consistent temporal input:

- All sequences are resized to a fixed temporal length.
- Videos shorter than the required sequence length are padded.
- Longer sequences are temporally sampled.

Data augmentation techniques including temporal scaling, time warping, and coordinate noise injection are applied during training to improve generalization.

2.2.3 Normalization

Joint coordinates are normalized before training to reduce positional variance across subjects and improve learning stability. Confidence scores associated with each keypoint are also included as additional input information.

2.3 ST-GCN++ Based Action Recognition

The core of the system is a fine-tuned ST-GCN++ architecture designed for skeleton-based action recognition.

2.3.1 Spatial Temporal Graph Modeling

The human skeleton is represented as a spatial-temporal graph:

- Spatial graph convolutions learn relationships between connected joints.
- Temporal convolutions model motion changes across video frames.

This allows the network to capture both body structure and movement dynamics simultaneously.

2.3.2 Network Architecture

The implemented model contains:

- Graph Convolutional (GCN) blocks
- Multi-Scale Temporal Convolution (MSTCN) modules
- Residual connections
- Batch normalization layers
- Adaptive average pooling
- Fully connected classification head

The network progressively learns hierarchical motion features through channel transitions from $64 \rightarrow 128 \rightarrow 256$ dimensions.

2.3.3 Transfer Learning and Fine-Tuning

A pretrained ST-GCN++ backbone is used to improve performance on the target dataset.

The pretrained weights are adapted and loaded into the custom implementation through transfer learning. Since the pretrained model was trained on a different skeleton format,

weight adaptation is performed for incompatible layers while preserving the majority of learned spatial-temporal features.

Training is performed in two phases:

1. Head Training Phase - the backbone is frozen while only the classifier head is trained.
2. Full Fine-Tuning Phase - the entire network is unfrozen and trained using a smaller learning rate.

This strategy improves convergence and prevents overfitting on the relatively small dataset.

2.4 Training Configuration

The model is trained using:

- Cross Entropy Loss with label smoothing
- Adam optimizer during head training
- SGD optimizer during full fine-tuning
- Cosine Annealing learning rate scheduling
- Weighted sampling to address class imbalance

Performance is evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix analysis

2.5 Dataset Description

Penn Action Dataset: <https://dreamdragon.github.io/PennAction/>

The proposed system uses the Penn Action dataset, which contains annotated human action videos with 2D body joint keypoints for each frame.

Actions used: pull-ups, push-ups, bench press, jumping jacks, sit-ups, jump rope, and squats.

The dataset provides skeleton sequences and action labels used to train and evaluate the ST-GCN++ model for exercise classification.

2.6 Real-Time Inference & Pose Detection

Human pose keypoints are extracted from video frames using MediaPipe, providing 2D joint coordinates with per-joint confidence scores. The system maintains a sliding window buffer of CLIP_LEN frames (configurable per dataset) with temporal padding for sequences shorter than required. Skeleton frames are validated using a confidence threshold of 0.40 for classification and 0.15 minimum ankle visibility to ensure temporal consistency before feeding into the model.

Real-time accuracy is delivered to users through an on-screen overlay displaying classification confidence percentages (0-100%) and session-level accuracy (computed against ground truth when provided) updated live during inference.

2.7 Biomechanical Feedback & Rep Counter

A rule-based form engine evaluates exercise technique by computing 8 joint angles (elbows, knees, hips, shoulders, wrist offsets) and comparing them against exercise-specific ideal ranges.

Rep counting uses state machines with per-exercise thresholds: squats (down 110°/up 160°), push-ups (down 85°/up 145°), bench press (down 0.25/up 0.70 normalized offset), pull-ups (down 145°/up 75° reversed), sit-ups (down 145°/up 70° reversed), jumping jacks (down 40°/up 130°), and jump rope (down 0.20/up 0.38 offset).

Form quality is scored 0-100% based on deviation from ideal angles at peak effort positions.

3. RESULTS & EVALUATION

The fine-tuned ST-GCN++ model achieved 84.93% validation accuracy on the exercise classification task, demonstrating balanced performance across all seven exercise classes. Session-level tracking compares predicted repetition counts against ground-truth values when available, enabling quantitative evaluation of counting performance. Additionally, confidence thresholding reduced false-positive predictions by approximately 40% compared to unthresholded outputs, improving overall prediction stability and reliability.

3.1 Comparison with Existing Biomechanical Analysis Tools

The compared systems represent different categories of biomechanical and motion-analysis tools, ranging from pose-estimation frameworks to professional motion-capture platforms and sports coaching applications. Kinexis was evaluated against several existing tools,

including **OpenPose**, **Kinect SDK**, **Vicon**, **Dartfish**, and **Coach's Eye**. While many existing systems specialize in either pose estimation, motion capture, or manual sports analysis, Kinexis combines **real-time exercise recognition**, **repetition counting**, and **automated kinematic feedback** into a *single integrated pipeline*.

Unlike *OpenPose*, which primarily focuses on pose estimation and outputs raw keypoint data, Kinexis provides exercise-specific analysis and feedback tailored to gym movements such as squats, push-ups, and pull-ups. Compared to *Kinect SDK*, which was mainly designed for gaming and general skeletal tracking, Kinexis offers more specialized exercise classification and form assessment capabilities.

Professional motion capture systems such as *Vicon* provide highly precise 3D tracking and real-time motion analysis, but they require expensive multi-camera setups and specialized environments. In contrast, Kinexis operates using a standard webcam and GPU, making it more accessible and suitable for consumer-level applications. Similarly, *Dartfish* and *Coach's Eye* support sports analysis through video annotation and playback tools, whereas Kinexis automates feedback generation using joint-angle analysis, form scoring, and repetition counting.

A major advantage of Kinexis is its **lightweight and extensible architecture**, which allows deployment on consumer hardware without specialized sensors. The system also provides **immediate corrective feedback during exercise execution**, enabling practical applications in fitness training, rehabilitation, and sports coaching. However, compared to professional biomechanical systems, Kinexis is *limited to 2D pose estimation*, *lacks kinetic measurements* such as force and muscle activation data, and *currently supports only seven exercise classes*. Its performance is also affected by camera positioning, occlusion, and lighting conditions.

Overall, Kinexis bridges the gap between expensive professional motion-capture systems and basic pose-estimation tools by delivering an **affordable, near real-time, and exercise-focused motion analysis platform**.

4. Architecture Diagram

** Diagram attached with submission in a separate .zip file.*